

SSD_Voice Intercom FAQ

1. Debugging process

- Set the computer IP and the development board in the same network segment
- Create a shared directory share, and mount it to the development board/mnt for grabbing the intermediate results of the algorithm, the IP is the IP set by yourself

```
mount -t cifs -o username="xxxxx",password="xxxxx"  
//192.168.1.203/share /mnt/
```

- View audio parameters (can only be viewed at runtime)

```
cat /proc/mi_modules/mi_ao/mi_aox  
  
cat /proc/mi_modules/mi_ai/mi_aix
```

- Grab the intermediate pcm, first set the AI/AO dump directory (you can write the following code into /customer/demo.sh)

```
export MI_AI_DUMP_PATH=xxx  
  
export MI_AO_DUMP_PATH=xxx
```

Then set the dump of the corresponding algorithm to 1

```
export MI_AI_DUMP_AEC_ENABLE=1  
  
export MI_AI_DUMP_VQE_ENABLE=1  
  
export MI_AO_DUMP_VQE_ENABLE=1
```

2. Common problems and solutions

The solutions to common problems are given here for reference only. Please refer to the actual situation during the debugging process.

problem appear	Phenomenon	reason	solution
There is a problem that the noise floor cannot be eliminated during calls	There is a large noise between the voices, and then it is suppressed	Persistent noise, but short speech interval, resulting in slow NR convergence	Comfort Noise Assisted NR Convergence with AEC on
	There is a little bit of echo between speeches ("squeaking noise")	The host AEC is not clean and leaves a tail sound	Grab the sound file analysis, if the frequency is fixed, use the AO end EQ/HPF to eliminate it. If the frequency is not fixed, use the AO side AGC, and set the noise gate to cut off the unclean echo of the host.
one-way intercom problem	obvious echo	The extension AEC is not clean	If the host hears an echo: the AEC of the extension is not clean, and the audio file of the extension is analyzed . 1. Adjust the frequency band division and elimination intensity of the AEC of the extension. 2. Reduce the gain of the extension mic/speaker

		3. Test the airtightness of the extension and eliminate the structural reasons 4. If the echo is a fixed frequency band, use the AI EQ to cut it 5. Reduce the mic gain of the host	
Host AEC is not clean		The extension hears an echo: the host AEC is not clean. 1. Reduce the gain of the mic/speaker of the main unit 2. Test the airtightness of the main unit and eliminate the structural reasons 3. Reduce the mic gain of the extension 4. If the echo from the main unit is a fixed frequency band, you can use the EQ on the AO end of the extension to cut	
Both sides talking at the same time	Loud and loud	The voice from the extension to the host is loud and quiet	抓出分机端的AI音档分析 1. 可能分机AEC开得太强，调整AEC强度 2. 若分机speaker端有开启AGC，则看是否为AGC参数没有调好，调整AGC参数 3. 若分机端的信号没问题，则可能为主机端speaker开启了AGC/EQ等算法，把信号处理出问题了，这里最好的处理方法是解主机。（从分机解需要分机去猜主机的行为）
主机到分机的声音忽大忽小		抓出分机AO端的音档分析。若主机过来的声音本来就是这样，则可能是主机mic端的AEC/AGC处理的问题。 1. 可能是主机AEC消得太干净。（这里可能尝试将主机mic/speaker、分机mic的增益减小来确认是否是主机AEC的问题） 2. 可能是主机AGC砍掉。（可以将主机的mic增益加大来实验，看是否为主机端AGC砍掉） 若主机过来的声音没有问题，则确认是否为分机AO端算法处理的问题还是分机结构/speaker的原因。 1. 若分机AO端有开启AGC，则检测是否为AGC参数没有调好导致的。	

		2. 可能为结构或speaker的原因导致某些频段信号打不出来
漏字 (全消 没了)	AEC开得太 强	1. 调整AEC强度 2. 减小mic/speaker端或对方mic的增益
	Speaker端有 打开AGC/EQ 等算法	调整相应的算法参数
响度小		1. 调整speaker端的增益或对方mic的增益 2. 结构问题
喇叭声音失真、破音、 颤音		1. 调整speaker端的增益 2. 调整对方的mic端的增益 3. 检查结构问题

注意：

- 通过分析中间抓取的PCM确定原因
- Determine the influence of the frequency band according to the spectrogram, calculate the frequency band where the noise occurs ($sr \cdot T / 128 = fs$, sr is the maximum frequency that can be obtained by sampling, T is the bin number, fs is the frequency corresponding to the bin), and then set the corresponding parameters, remove the corresponding frequency band
- Turn off all algorithms to determine the cause of audio problems
- Blocking the board's microphone to determine the cause of the audio problem
- Turn off the speakers to determine the cause of the audio problem
- When it comes to algorithm updates, it is recommended to back up the original algorithm library first.